

When a Fixed-Batch Risk Estimate Changes an Obedience Test

Research phase: DRAFT after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

Abstract

The supplied pair consists of a paper on rewards for AI agents and a local paper on fixed-batch risk estimates in a resource-allocation game. We asked whether an estimated risk value can change the first paper’s test for supporting a signal-dependent action policy with rewards that do not depend on the state. For every finite batch size, the peers found a two-signal example that passes the test under the expected empirical estimate and fails it under true conditional risk. They also checked a margin test and a common-reward test for bounded value errors. The verifier accepted these narrow results but left the thread in DRAFT because the report stage under a specified risk timing, realised batches, and a broader novelty claim remain open.

1 Introduction

The supplied Q, by Bergemann, Koh and Morris, studies how to reward an AI agent when its preferences and abilities are private. The agent may report a type and then choose an action after receiving a signal about the world. Q asks when rewards can make both the report and the action follow a desired policy. In the special case where an action’s reward cannot depend on the state, Q gives a test based on cycles of signals: the gains from following the policy around every such cycle must add to at least zero [1].

The supplied P studies a different setting: a quadratic resource-allocation tax game evaluated with conditional value-at-risk, or CVaR. CVaR measures the average loss in a specified upper tail. P separates true CVaR from the average empirical CVaR produced by repeatedly drawing a fixed-size batch. Its one-sample example has an exact equilibrium for the empirical objective that fails participation under true risk. Its value-error bounds concern the expected empirical objective, subject to its bounded-loss and valid-query assumptions [3].

P calls its resource-allocation source “Q”, but that is a different paper from the Q supplied here. The peers therefore pursued a new, narrower connection: could P’s sampling bias change Q’s signal-cycle answer? They treated negative *conditional* CVaR, evaluated after a signal, as the agent’s action utility in Q. This is an explicit modelling choice, not a result about the tax game in P or about CVaR of an entire plan before the signal.

2 What was done

The peers first checked the identity of the papers and set aside a direct transfer of P's tax results. They then reduced Q to one agent type, two equally likely states that the signal reveals perfectly, and two actions, a and b . They let a fresh batch estimate the risk of each action after the signal. Ada proposed the reversed-risk example; Emmy checked it against Q's cycle condition and extended it from one observation to every finite batch size. Both peers checked the reward inequalities independently. Ada then derived a test for a reward that works across a box of possible utility errors, and Emmy checked it. Emmy separately tested what happens if the agent instead assesses a whole plan by CVaR before seeing the signal.

For the main example, fix a positive loss scale h and a finite batch size m . A risky action loses either 0 or $2h$, each with probability one half. A safe action has constant loss c_m . In state 0, a is risky and b is safe; in state 1, their roles reverse. The proposed policy selects the risky action in both states. Let q_m be the expected empirical upper-half CVaR of the risky loss from m independent observations. An all-zero batch has probability 2^{-m} , so its empirical CVaR is zero; every other empirical value is at most $2h$. The peers chose

$$q_m \leq 2h(1 - 2^{-m}) < c_m := 2h - h2^{-m} < 2h.$$

The true upper-half CVaR of the risky action is $2h$. These inequalities put the safe loss between the estimated and true values of the risky loss.

3 Results

Under expected empirical CVaR, the proposed risky action has lower estimated loss than the safe one after either signal. A reward of zero for each action therefore supports the policy. Under true conditional CVaR, let $r(a)$ and $r(b)$ be rewards that depend only on the action. Obedience in state 0 requires $r(a) - r(b) \geq 2h - c_m$; obedience in state 1 requires $r(b) - r(a) \geq 2h - c_m$. Since $2h - c_m = h2^{-m} > 0$, both requirements cannot hold. Equivalently, Q's two-signal cycle has a positive estimated sum and the true sum $-2h2^{-m}$. Thus, for every finite m , some instance passes the expected empirical test but no state-independent reward supports the same policy under true conditional risk. The safe loss c_m varies with m ; this is not failure for one fixed instance as the batch grows. Ada's derivation in `ada/finite_batch_implementability.md` and Emmy's check in `emmy/findings.md` support this result.

The peers also checked when value accuracy protects the cycle test. Write $U(x, s)$ for true utility of action x after signal s , and $\tilde{U}(x, s)$ for its estimated utility. If $|U(x, s) - \tilde{U}(x, s)| \leq e$ for every compared action and signal, each term of a signal cycle can change by at most $2e$. Hence an estimated cycle of length K with a margin of at least $2Ke$ still meets Q's true cycle condition. If all relevant simple cycles have that margin, Q's result supplies some supporting state-independent reward; the estimated reward itself need not work under true utility.

A stronger question asks for *one* reward that works for every utility array in an independent error box of width e . Let B be the actions prescribed somewhere by the policy, and let $a(s)$ be its action after signal s . For distinct $x, y \in B$, the peers assigned the directed edge x to y the weight

$$w(x, y) = \max_{s: a(s)=x} \{\tilde{U}(y, s) - \tilde{U}(x, s) + 2e\}.$$

Their check showed that a common state-independent reward exists exactly when no directed cycle of these prescribed actions has positive total weight. The reason is that robust obedience says $r(x) - r(y) \geq w(x, y)$. Summing those inequalities rules out a positive cycle, while Q's

path construction gives rewards when none exists. Actions never prescribed can be prohibited. This exact statement concerns the independent error box; using P's bound for a particular application still requires P's assumptions.

4 Objections and limits

The main example uses risk assessed after each signal. Emmy checked a separate timing objection: an agent may instead rank a whole signal-dependent plan by CVaR of the joint signal-and-loss distribution before seeing a signal. In a two-signal example in the ledger, averaging conditional CVaRs favoured one report, while ex-ante CVaR favoured another. Thus, averaging local risk values does not generally give Q's report-stage continuation value. More generally, if a loss J lies between 0 and M , S is the signal, and β is upper-tail mass, the peers established

$$0 \leq \text{CVaR}_\beta(J) - \mathbb{E}[\text{CVaR}_\beta(J | S)] \leq (1 - \beta)M.$$

This difference can remain with exact risk values. The ledger cites Miller and Yang [2] and Rudloff, Street and Valladão [4] on the known dynamic-CVaR timing issue. Its limited searches did not settle novelty.

Q also allows state-dependent rewards outside the subclass studied here. The example does not rule those out. It concerns the *expected* empirical value, not a guarantee for one realised batch or convergence of strategic learning. The peers established a two-signal obstruction and value-error tests, not a full theorem combining Q's action and type-report cycles under CVaR.

5 Why the thread stopped

The verifier's single round accepted the finite-batch reversal, the reward inequalities, and the value-error graph test. It recorded DRAFT because the supported conclusion is narrow: the peers had not fixed a risk-timing model for the report stage or proved the corresponding nested-cycle result. The smallest next step is to specify whether reports are evaluated by ex-ante plan CVaR or by conditional CVaR after each signal, then solve a two-type, two-signal report game under that choice. A realised-batch guarantee and a fuller prior-work check would remain separate work.

References

- [1] Dirk Bergemann, Andrew Koh, and Stephen Morris. Mechanism design for alignment and control, September 2026. URL: <https://arxiv.org/abs/2609.01595>, arXiv:2609.01595.
- [2] Miller and Yang. arXiv:1512.05015. URL: <https://arxiv.org/abs/1512.05015>.
- [3] Pathfinder research campaign. Fixed-batch CVaR and participation in a quadratic resource-allocation mechanism, September 2026. Local manuscript, pathfinder-local-S4.
- [4] Rudloff, Street, and Valladão. Optimization online 2010/12/2860. URL: <https://optimization-online.org/2010/12/2860/>.